Clinical Toxicology

ISSN: 1556-3650 (Print) 1556-9519 (Online) Journal homepage: http://www.tandfonline.com/loi/ictx20

Fatal poisoning by terbufos following occupational exposure

Yue Liang, Fang Tong, Lin Zhang, Wenhe Li, Weisheng Huang & Yiwu Zhou

CLINICAL TOXICOLOGY, 2017
https://doi.org/10.1080/15563650.2017.1340647

Taylor & Francis
Taylor & Francis Group

SHORT COMMUNICATION

Fatal poisoning by terbufos following occupational exposure

Yue Liang[a], Fang Tong[a], Lin Zhang[a], Wenhe Li[b], Weisheng Huang[a] and Yiwu Zhou[a]

[a]Department of Forensic Medicine, Huazhong University of Science and Technology, Tongji Medical College, Wuhan, PR China; [b]Department of Pathology, The Central Hospital of Wuhan, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, PR China

ABSTRACT

Context: Terbufos (TBF) is a class Ia (extremely hazardous) organophosphate pesticide (OP) and its distribution in industrialized countries has been severely restricted. Thus, acute occupational poisoning is rather uncommon. However, it still occurs in rural areas of some developing countries, where the sale of TBF is not controlled and its use is thus not properly regulated. We report a case of a 43-year-old female farmer who died after applying TBF granules.

Case: The patient died within 3 h after applying 20 bags of 5% TBF granules (900 g per bag). Investigation showed that her personal protective equipment (PPE) did not provide effective protection against dermal and inhalational exposure. Postmortem analysis revealed extremely low red blood cell acetylcholinesterase activity. Toxicological analysis of TBF showed 1.45×10^{-2} μg/ml in the heart blood and 0.17 μg/g in the liver.

Discussions: This patient died as a result of toxicity from dermal and inhalational exposure to TBF. Over-application, improper equipment, inadequate and defective PPE, and lack of hygienic precautions were all contributing factors.

Conclusions: TBF is a highly toxic OP. Inadequate regulatory control, improper environmental application, and ineffective PPE resulted in a fatal human exposure.

ARTICLE HISTORY
Received 1 December 2016
Revised 3 June 2017
Accepted 5 June 2017
Published online 4 July 2017

KEYWORDS
Organophosphate pesticides; terbufos; fatality; occupational exposure

Introduction

Terbufos (TBF) [S-*t*-butylthiomethyl-O,O-diethyl phosphorodithioate], an organophosphate pesticides (OPs), is extensively used worldwide as an insecticide and nematicide [1]. The toxicity is due to cytochrome P450-dependent oxidative desulfuration to form its O-analog metabolites, terbufos-oxon. This active form inhibits acetylcholinesterase (AChE) irreversibly in the central and peripheral nervous systems leading to an accumulation of acetylcholine and cholinergic crisis [2].

TBF is a WHO class Ia OP and its distribution in industrialized countries has been severely restricted. Thus, acute occupational poisoning is uncommon [3,4]. However, it still occurs in rural areas of some developing countries, where the sale of TBF is not controlled and its use is thus not properly regulated. We report a case of fatal human poisoning with TBF.

Case report

A 43-year-old female farmer was transported to the state emergency department (ED) following a 3-h application of TBF. She was suffering from a headache, dizziness, tightness of chest, dyspnea, and progressive diaphoresis accompanied by nausea, vomiting, and an altered level of consciousness. Upon arrival (approximately 30 min later), she became unconscious and suffered a respiratory arrest. On physical examination, she was pale, salivating, and diaphoretic, with miotic, non-responsive pupils, and with an intense odor of pesticide emanating from her clothing. Vital signs were undetectable. She was pronounced dead 1 h later despite intensive resuscitation efforts.

Further history indicated that the patient's main task that morning had been mixing and scattering TBF pesticide granules, while her co-workers applied a pesticide-free fertilizer. The TBF was in the form of 5% (active ingredient) granules (900 g per bag) with a violet aposematic coloration and it was her first time using it. Twenty bags were mixed with coarse sand and scattered on fields of peanuts (an area of about 0.33 hectares) manually. She complained about the pungent odor during the application. After 3 h, she developed a headache, dizziness, tightness of chest, nausea and began to vomit and returned home. The other workers did not suffer any unusual symptoms. When discovered, some 30 min later, she was noted to be lethargic and was then transported to the ED. Per report, as personal protective equipment (PPE), the patient had been using a bandana as an improvised "mask", torn rubber gloves and boots. Additionally, she had been wearing a short-sleeved shirt because of the warm weather, about 30 °C.

Post-mortem, toxicological, and laboratory findings

Violet TBF powder was found on her right fingers and fingernails (Figure 1). Internal examination revealed foamy material

CONTACT Yiwu Zhou zhouyiwu@outlook.com Department of Forensic Medicine, Huazhong University of Science and Technology, Tongji Medical College, No. 13 Hangkong Road, Hankou, Wuhan 430030, PR China

Figure 1. Violet powder could be seen on her right fingers and fingernails.

in the respiratory tract. Pathologic findings included moderate cerebral edema, mild neuronal degeneration/necrosis, myocardial contraction band necrosis, pulmonary edema and hemorrhage, focal hepatic necrosis, vacuolar renal tubular epithelium degeneration, and pancreatic necrosis and hemorrhage.

The heart blood, liver, stomach tissue, and samples collected from right fingers were preserved during autopsy and stored at −20 °C until analysis. TBF and malathion (internal standard) were supplied by the National Institute for Agrochemicals Control (Beijing, China). All the toxicological analyses were performed by the Department of Forensic Medicine of Tongji Medical College (Wuhan, China) using GC/MS (Agilent 6890 A GC coupled with a MSD 5975 B).

Analytical data revealed 1.45×10^{-2} μg/ml TBF in the blood and 0.17 μg/g in the liver. No TBF was found in the stomach. The samples collected from fingers revealed the presence of TBF. No other compounds were detected.

Red blood cell (RBC) AChE activity on post-mortem blood was 0.3k U/L (normal range 3.5–8.5 kU/L).

Discussion

OPs can enter the human body through dermal absorption, inhalation, or ingestion [5]. Poisoning episodes by dermal and respiratory exposure usually occur among farm workers during spraying, mixing, and diluting pesticides. The use of defective PPE is an important contributor to poisoning [6]. In our case, improper or defective PPE did not provide effective protection against dermal exposure as evidenced by the detection of TBF on her skin and evidence of systemic toxicity.

The dermal absorption rate is determined by the specific OPs type, skin condition, and contact site/duration [5]. TBF is highly toxic and the dermal LD50 ranges from 7 to 10 mg/kg in rats equivalent to 1.1–1.6 mg/kg in humans [3,7]. In this case, fatality should be expected after an absorption dose of 60–100 mg, 0.006–0.01% of the actually applied dose (0.9 kg, 20 bags of 5% granules with net content 900 g per bag). Although the granular formulation is less readily absorbed [8], such a high dose appeared to have been achieved here due to the following factors [5,8,9]: (i) temperature and humidity were high, increasing the dermal absorption rate secondary to sweating and increased blood circulation; (ii) absorption continued as TBF remained in contact with the skin on her hands and clothing; and (iii) defective PPE allowed dermal exposure. It is possible that TBF residues were transferred inadvertently from her hands to other parts of the body where the absorption rate is higher, such as the eyes and forehead.

The high volatility of TBF used on dry soil created a potential risk for respiratory exposure [10] and the bandana used by the victim provided inadequate respiratory protection. Furthermore, there was over-application of TBF in that a total of 18 kg product were used on approximate 0.33 hectares, corresponding to around 55 kg/ha. This is higher than the recommended use rate (36–45 kg/ha) for most crops, and which would increase the air concentration of TBF. Ingestion was excluded by the absence of TBF in the GI tract.

Death from acute TBF poisoning is caused by an acute cholinergic crisis (ACC) due to irreversible inhibition of AChE leading to acetylcholine accumulation at the peripheral muscarinic and nicotinic synapses and in the central nervous system (CNS). The onset of ACC varies from few minutes to several hours post-exposure [1,11]. In this case, the clinical course corresponded to ACC with muscarinic (diaphoresis, vomiting, miosis, salivation), nicotinic (pallor, muscle weakness with respiratory failure), and CNS (headache, dizziness, altered level of consciousness) symptoms [12], which occurred within hours post-exposure. The severe reduction of RBC AChE activity confirms exposure with systemic toxicity. Management of OP toxicity includes early recognition, rapid decontamination, and treatment with atropine and oximes [11].

Conclusions

TBF is a highly toxic OP. Inadequate regulatory control, improper environmental application, and ineffective PPE resulted in a fatal human exposure.

Acknowledgements

We thank Dr. Basil Arif for editing the manuscript.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was supported by National Key Research and Development Program of China (No. 2016YFC0800701), the Fundamental Research Funds for the Central Universities, HUST (No. 2016JCTD117), and the National Natural Science Foundation of China (No. 81273336).

References

[1] Lorke DE, Nurulain SM, Hasan MY, et al. Prophylactic administration of non-organophosphate cholinesterase inhibitors before acute exposure to organophosphates: assessment using terbufos sulfone. J Appl Toxicol. 2014;34:1096–1103.

[2] Choung CB, Hyne RV, Stevens MM, et al. Toxicity of the insecticide terbufos, its oxidation metabolites, and the herbicide atrazine in binary mixtures to *Ceriodaphnia cf dubia*. Arch Environ Contam Toxicol. 2011;60:417–425.

[3] World Health Organisation. The WHO recommended classification of pesticides by hazard and guidelines to classification 2004. Geneva: International Programme on Chemical Safety; 2005.

[4] Peter JV, Jerobin J, Nair A, et al. Is there a relationship between the WHO hazard classification of organophosphate pesticide and outcomes in suicidal human poisoning with commercial organophosphate formulations? Regul Toxicol Pharmacol. 2010;57:99–102.

[5] Schulze LD, Ogg CL, Vitzthum EF. Signs and symptoms of pesticide poisoning. University of Nebraska Cooperative Extension. EC97-2505-A; 1997.

[6] Palis FG, Flor RJ, Warburton H, et al. Our farmers at risk: behaviour and belief system in pesticide safety. J Public Health. 2006;28:43–48.

[7] Nair AB, Jacob S. A simple practice guide for dose conversion between animals and human. J Basic Clin Pharm. 2016;7:27–31.

[8] Macfarlane E, Carey R, Keegel T, et al. Dermal exposure associated with occupational end use of pesticides and the role of protective measures. Saf Health Work. 2013;4:136–141.

[9] Oesterlund AH, Thomsen JF, Sekimpi DK, et al. Pesticide knowledge, practice and attitude and how it affects the health of small-scale farmers in Uganda: a cross-sectional study. Afr Health Sci. 2014;14:420–433.

[10] Fuller BW, Landsman TM, Boetel MA, et al. Soil-surface collection of volatilized insecticides in relation to corn rootworm management. J Agric Urban Entomol. 1997;14:339–407.

[11] Chowdhary S, Bhattacharyya R, Banerjee D. Acute organophosphorus poisoning. Clin Chim Acta. 2014;431:66–76.

[12] Paudyal BP. Organophosphorus poisoning. J Nepal Med Assoc. 2008;47:2518.